

$$Z\text{-sum} = 16$$

No. as not all potentials are strictly positive

Belief Propagation

a) probabilities be inferred by Markovian chain from neighbors

$$b) \mu_{f_2 \rightarrow b}(b) = \sum_{c,d} f_2(b,c,d) \cdot f_3(c) \cdot \mu_{d \rightarrow f_2} = b + 0.8$$

$$\mu_{f_2 \rightarrow c}(c) = \sum_{b,d} f_2(b,c,d) \cdot \mu_{d \rightarrow f_2}(d) \cdot \mu_{b \rightarrow f_2}(b) = 1.2c + 1.4$$

$$\mu_{f_2 \rightarrow d}(d) = \sum_{b,c} f_2(b,c,d) \cdot \mu_{b \rightarrow f_2}(b) \cdot \mu_{c \rightarrow f_2}(c) = \sum_{b,c} f_2(b,c,d) \cdot f_3(c) = 0.4d + 1.1$$

$$p(a) \Big|_{a=0} = \frac{8}{26} \quad p(a) \Big|_{a=1} = \frac{18}{26} \quad p(b) \Big|_{b=0} = \frac{8}{26} \quad p(b) \Big|_{b=1} = \frac{18}{26}$$

$$p(c) \Big|_{c=0} = 0 \quad p(c) \Big|_{c=1} = 1 \quad p(d) \Big|_{d=0} = \frac{11}{26} \quad p(d) \Big|_{d=1} = \frac{15}{26}$$

c) ① - f_1 - ② - f_2 - ③ - f_3

$$\mu_{f_1 \rightarrow b}(b) = \mu_{b \rightarrow f_2}(b) = \max_a f_1(a,b) \cdot \mu_{a \rightarrow f_1}(a) = \begin{cases} 0.5 & b=0 \\ 0.3 & b=1 \end{cases}$$

$$\mu_{f_2 \rightarrow c}(c) = \mu_{c \rightarrow f_3}(c) = \max_b f_2(b,c) \cdot \mu_{b \rightarrow f_2}(b) = \begin{cases} 0.1 & c=0, b=0 \\ 0.2 & c=1, b=0 \end{cases}$$

$$\mu_{f_2 \rightarrow b}(b) = \mu_{b \rightarrow f_1}(b) = \max_c f_2(b,c) \cdot f_3(c) = \begin{cases} 0.2 & b=0 \\ 0.45 & b=1 \end{cases}$$

$$\mu_{f_1 \rightarrow a}(a) = \max_b f_1(a,b) \cdot \mu_{b \rightarrow f_1}(b) = \begin{cases} 0.1 & a=0 \\ 0.045 & a=1 \end{cases}$$

$$a^* = \arg \max \mu_{f_1 \rightarrow a}(a) = 0$$

$$b^* = \arg \max \mu_{f_1 \rightarrow b}(b) \cdot \mu_{f_2 \rightarrow b}(b) = 0$$

$$c^* = \arg \max \mu_{f_2 \rightarrow c}(c) \cdot \mu_{f_3 \rightarrow c}(c) = 1$$